Eco Solutions Innovations Ltd

Start-Up Business plan

Venture team:

Group members' names and student number
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Unit: BMO2005 Innovation and Entrepreneurship

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Key details

Registration details

Business name	Eco-Solutions Innovations Ltd.
Date registered	14/01/2024
Business structure	Company

Key contact details

Name	Lakmal Sampath
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The start-up

Plan summary

Our business	Eco-Solutions Innovations Ltd. is committed to addressing the widespread problem of laundry microfiber pollution,a significant environmental concern.Our innovative solution,the Green Fiber Filter Sphere (G2FS),is designed to combat this issue and mitigate the harm caused by plastic fibers in laundry detergents to ecosystems.Scientific predictions,as highlighted by Sewwandi (2021), indicate a concerning future where a significant percentage of seabirds and sea turtles will have consumed plastics by 2050. Microfibers,a new contaminant in marine environments by Liu (2021),are predominantly sourced from textiles.The G2FS serves as a groundbreaking solution,dedicated to collecting and preventing microfiber contamination within laundry cycles. Eco-Solutions Innovations Ltd. envisions a future where each laundry wash contributes to a healthier planet.The G2FS, a crucial component in addressing one of today's most pressing environmental challenges,reflects our commitment to environmental conservation and sustainable practices.
Our target market	G2FS is aimed at a target audience including sustainability-minded households residing near water bodies,emphasizing the product's alignment with their environmental goals.Eco-friendly businesses such as dry cleaners,laundry services,and lodging facilities are crucial clients seeking to enhance their eco-friendly offerings.Environmental organizations,governmental entities and educational institutions to promote cleaner habits.Eco-aware internet platforms,as well as retailers and distributors specializing in sustainable products,constitute significant markets.

Our value propositions	G2FS has an efficient microfiber gathering and prevention mechanism contribute to cleaner water systems,safeguarding aquatic environments,and protecting marine life.The user-friendly design makes it accessible for individual customers,households and enterprises,catering to those seeking convenient ways to reduce their environmental footprint. Distinguishing itself from competitors like Cora Ball and Guppy Friend,G2FS boasts a triple filtration process,a significant advancement from single-layer designs.This triple-layered strategy ensures thorough trapping of microfibers during laundering, offering users a higher level of environmental protection. (Refer Appendix D) Key differentiators of G2FS is self-cleaning technology,eliminating the need for manual maintenance ensures consistent peak performance over time,enhancing user convenience and it incorporates a smart sensor technology allowing the filter to adapt to various laundry conditions,optimizing its effectiveness based on variables such as detergent type and water temperature.
Innovation	G2FS redefines microfiber filtration capabilities with its triple filtering mechanism,self-cleaning technology and smart sensor integration.During laundry cycles,the triple-layered filter system effectively reduces microfiber pollution,and the self-cleaning technology guarantees little maintenance.

Our why

Eco-Solutions Innovations Ltd. was based on a belief in the transformative power of individual actions and personal commitment to environmental stewardship.Investing personal funds in the company underscores a dedication to environmental preservation and sustainable development.The core motivation is the desire to minimize plastic and microfiber release during washing cycles,thereby preventing ocean pollution.Our commitment extends to the preservation of marine life and seabird species, aligning with a broader mission to support the well-being and balance of Earth's ecosystems.

Our vision

With cutting-edge products such as G2FS,our company promotes sustainable living throughout the world with the goal of encouraging eco-friendly choices and establishing sustainable living as a means of life.

Our mission

Our company is empowering people and businesses worldwide by using creative solutions to revolutionize sustainable living. Our goal is to provide innovative,environmentally friendly products—like the Green Fiber Filter Sphere—that not only lessen their negative effects on the environment but also spur long-lasting change. Our goal is to ignite a world whereby conscious decisions guide daily actions, promoting a balanced coexistence of people and the environment.

Operations

Products and services

Product/service	Description	Unit price (incl. GST)
Green Fiber Filter Sphere (G2FS)	Gathers microfibers during laundry cycles with the triple filtering system to contribute to a cleaner environment.Its self-cleaning technology and user-friendly design ensure easy maintenance. For environmentally conscious customers,the Green Fiber Filter Sphere offers an affordable and sustainable option,conveniently accessible on platforms.	LKR 4200

Scope of products/services

The G2FS offers a solution to the global issue of laundry microfiber contamination.Catering to environmentally conscious customers,businesses,and institutions,it’ objective is to protect marine life and enhance waterway cleanliness by providing a practical and easily accessible tool that reduces microfiber discharge during laundry cycles.The product's global accessibility aligns with our commitment to promoting sustainable practices on a global scale.

Production

The production of the G2FS follows a year-long process to bring the innovation to the market. The initial five months are dedicated to research and development,involving data collection through surveys and forums.Simultaneously, testing is conducted to ensure the effectiveness of the product.

The production phase,spanning six months,includes outsourcing critical components such as the automated self-cleaning mechanism,nanofiber material,multi-filtration layers,anti-microbial coating,and smart sensor technology.Upon receipt from trusted suppliers,inspections are conducted to ensure adherence to required standards.Assembly takes place at our facility in Sri Lanka,utilizing advanced Mitsubishi electric instruments and a combination of manual labour processes.This assembly process,including packaging,spans approximately a month.

The packaging process involves outsourcing the box production,followed by manual wrapping to reduce costs and minimize defects that may go unnoticed by machines.Final quality inspections are conducted concurrently with the packaging process to guarantee the product's adherence to the highest standards.

Digital technology

Digital prototyping and simulation tools will enhance the accuracy of product design during the R&D phase,ensuring an efficient testing procedure.Supply chain management technologies will be employed to track and oversee the procurement of internal components,optimizing production schedules.In the assembly process,collaborative robots,or cobots will be deployed to enhance accuracy and productivity.Smart sensors and Internet of Things (IoT) technology will be integrated into the product,enabling adaptive functionality.The global online presence of the G2FS will be facilitated through e-commerce platforms like Amazon and Daraz.

To manage individualized consumer interactions across retail and online channels, a Customer Relationship Management (CRM) system will be implemented.Analytical software will be utilized to gather market insights, guiding ongoing product innovations.

Key resources

The key financial resources for our start-up are anchored in a 2Mn LKR initial capital investment, representing a crucial foundation for a successful launch and subsequent growth.This financial infusion will drive the initial phases of research,development,and manufacturing for the Green Fiber Filter Sphere.

Eco-Solutions Innovations' success is attributed to its strong human resources too,including innovative leadership from co-founders and a key collaborator-the technical developer.The team, including office assistants, managers, security guards, and cleaners, ensures effective operations and a diverse approach to essential resources, driving the startup towards sustainable and profitable operations.

Key activities

We commence with research and development, gathering crucial insights through forums and surveys that form the foundation for the G2FS.The subsequent stages involve product design and prototype development,aiming for a creative,effective, and sustainable solution.Our commitment to education involves informing the public about our program and the environmental impact of our product.Supply chain management and material sourcing play a crucial role,ensuring the acquisition of premium components. Consistent customer care and a robust feedback cycle constitute our primary operations,creating a customer-centric mindset.

Equipment

Eco-Solutions Innovations places great emphasis on sourcing high-quality components for the G2FS from internationally renowned suppliers. Elmarco(Czech Republic) supplying nanofiber materials,utilizing cutting-edge technology within the filtration system. Freudenberg(Germany),recognized for its innovative approach to enhancing product performance,supplies the multi-filtration layers.BioCote(UK) specializes in antimicrobial technology,providing a crucial coating for additional hygienic practices.

STMicroelectronics(Switzerland)a leading semiconductor solutions,contributes the smart sensor technologies essential for adaptive functionality.The automated self-cleaning system is sourced from Siemens AG(Germany),a multinational technology and automation leader,Mitsubishi's electronic equipment,selected for assembly,and packaging supplies from Neenah Papers(USA). Demonstrating our commitment to ecological sustainability,the GFS exterior is crafted from a ball material that feels like cotton.

Key people

Our staff

- **Founders:**
    - Lakmal Sampath
    - Thamasha Perera
    - Harini Perera
    - Anuradhi Ranasinghe
- **Packaging Team (8 workers)**
- **Production Team (3 workers)**
- **Security Team (2 Guards)**
- **Cleaning Staff (3 workers)**
- **Management Team:**
    - Production Manager
    - Human Resources Manager
    - Finance Manager
    - Marketing Manager
    - Logistics Manager
    - Customer Relations

Team structure

Guiding our startup's initiatives are our founding members and Supporting our creative founders is a talented staff of 29 members,functioning as an integrated team to drive our operations forward.

Within our team,eight members are dedicated to meticulous packaging,three handle production, with two overseeing quality control.The leadership team,comprising roles such as production manager, financial manager, purchasing expert, marketing specialist, logistics coordinator,and customer relations specialist, contributes to the strength and direction of our startup.Additionally we have three dedicated cleaning staff members and two security guards to ensure a secure and safe working environment in our manufacturing facility and office spaces.

Sources of advice and support

Support type	Name	Skills and experience
Mentors	Ms. Nilusha Gallage	Experiences in innovation and entrepreneurship, quality mentoring to help us to do the work in a quality manner. Moreover, deep experience and skills in the innovation and entrepreneurship industry.
Mentors	Ms. Bhasuri Amaratunga	Experience in the field of Human resources management and innovation and entrepreneurship and deep understanding of how the industry works.
Technical Developer	Mr. Thejana Asel Silva	Development of 3D models, prototypes and self-made electronical equipment's. Technical Expertise.

Key partnerships

As the previous statements clearly specifies, we collaborate with;

- Elmarco
- Freudenberg
- BioCote
- STMicroelectronics
- Siemens AG
- Mitsubishi Electronics
- Neenah Papers
- And Aitken Spence Pvt Ltd, as logistics partner managing product delivery and logistics,facilitating our global operations.
- Strategic retail partnerships with Keells, Daraz, Ikea outlets, and Amazon contribute to global product distribution.

The Market

Growth plan

The company's global focus mainly contributes for the increased business growth and sales while increasing customer interest is done by high standards and consistent quality.Collaborations with waste management companies and local retailers and highlighting the sustainability aspect of G2FS through educating customers,(as this product is lies in the introductory stage in the Sri Lankan market) will be done.Furthermore,we have set clear SMART objectives to grow in the industry. For example, to grow the sales;

Specific – Increasing 15% sales of G2FS annually.

Measurable – Generating a revenue of 46,200,000 LKR in the first year and achieve an annual growth of 15% each subsequent years through enhanced marketing strategies that boost the quantity of units sold.

Achievable – Conducting targeted marketing campaigns to reach the product to a wider audience including B2B marketing.

Relevance – Maintaining relevance to customer needs by introducing promotions and discounts to attract new customers and re-order rate by customers.

Timeliness – Achieving 15% sales increase at the end of each accounting period while monitoring the progress quarterly.

Promotional channels

Direct-to-consumer channels including local and international retail stores such as Keells Super and IKEA India,via official online platforms such as through the company's website, which provides the product details and the reviews,"Daraz", "Amazon", "eBay" and "bed bath & beyond" ,through **e-mail and SMS marketing** and utilising **social media** such as Facebook, Instagram and Twitter to enhance the visibility,conduct promotions/giveaways, interact with customers directly).Furthermore,reaching customers by hosting **eco-friendly trade exhibitions** to educate and create awareness about G2FS by conducting **informative workshops** and attract the potential consumer-base as well as the potential collaborating business partners.

Collaborating with influencers and environmental organisations for in-store presentations to engage the target market, also with the national projects organised for cleaner oceans and engage in **cooperative marketing campaigns** with eco-conscious subscription box providers encouraging to include the G2FS in their offerings.

Traction

By conducting online surveys, social media questionnaires and meeting eco-conscious customers for their ideas and feedback to ensure the desirability and demand (Barringer & Ireland, 2022) for the G2FS. Furthermore, affirming the industry-attractiveness by identifying that the sector to which G2FS belongs is an emerging industry with opportunities for expansion. Also, the target market attractiveness for G2FS can be recognised by (National Cleaner Production Centre, 2018), which depicts a 79% of Sri Lankans are keen in sustainable products. Another contributing factor for potential customers to have interest in G2FS is the availability of organisational resources, skills and expertise of the employees in the specialised area of microfiber filtration.

Industry

The G2FS product falls within the "Water Filtration Industry," which aims to eliminate impurities and contaminants from water,addressing water pollution.Various technologies, such as sand-filters and bio-filtrations,are employed in this sector.This industry significantly impacts public health by ensuring water safety and preventing waterborne diseases.It is a burgeoning sector with global demand from both developed and developing nations,driven by increased awareness of water quality and environmental sustainability.Ongoing research and development,coupled with industry innovation,further propel its growth.

The Water Filtration Industry is predicted to experience a 7% market size increase by 2028(Services, 2023).Current industry trends involve the integration of smart-filtration systems through IoT,enhancing efficiency with advanced materials like membrane filtration and nanotechnology and the adoption of sustainable water filtration solutions, such as G2FS.

The competition

Competitor name	Dropps	Guppy Friend	Cora ball
Strengths	High quality and lower prices	Lower material costs have expanded the business for areas such as sink filters and cleaning clothes (GUPPYFRIEND, 2024).	Attractive design inspired by the corals, prevents the microfiber shedding from laundry
Weaknesses	Doesn’t have an exact product for microfiber pollution	Doesn’t integrate advanced technology for their production process.	Not suitable for every clothing type,Focuses the high-end market
What we’ll do differently	We hope to differentiate our product to the dryer ball segment	We hope to differentiate our product to other categories, and grow firmly	We present affordable yet high quality microfibre filtration balls with three multi-filtration layers,that caters to every type of clothing,ensuring efficiency.

Pricing strategy

After considering the cost of the materials and labour,pricing strategies of the competitors and the value that the product offers to the consumer,we have set the price for LKR 4,200,which will be converted based on the ongoing exchange rate in the event of international sales.In the domestic market,we implement price skimming strategy, while in the international market, we adopt the price penetration strategy.Furthermore,with the market competition and trends,we will remain flexible with pricing,making slight adjustments when necessary.

Distribution strategy

We leverage various channels, including our company website, retail supermarket chains like Keels Super, and online platforms such as Daraz and eBay, to offer our product locally. Additionally, we extend our reach globally by exporting to IKEA stores in India and utilizing platforms like Amazon.Our in-house packaging operations,product storage, and distribution are seamlessly managed through collaboration with Aitken Spence Cargo Pvt Ltd.This partnership ensures efficient delivery,reduced lead times, and heightened responsiveness throughout the distribution process.

SWOT analysis

Strengths	Utilisation of innovative materials and components for G2FS,our product has an extended lifespan,reducing the frequency of replacements,our commitment on continuous innovation can be considered as a strength,as we aim to develop a more efficient yet superior quality product.
Weaknesses	Higher manufacturing costs including high material and labour costs, Difficulties in adopting to advanced technologies right-away,as a new entrant to the market,there might be limitations in establishing a stronger brand.

Opportunities	Since the market continues to expand in Sri Lanka, it allows to differentiate G2FS in the market and to standout. Having a loyal consumer-base,opportunity to collaborate with hotel-chains,environmental companies to expand distribution and improve credibility.
Threats	Possibility of increased costs and reduced profitability due to the fluctuations in economic factors,Lack of awareness about microfibre filtration balls and their purpose,an intense competition exists in the global market which might negatively impact our product.

Weaknesses - High manufacturing costs are mitigated through process automation,reducing labour expenses.Diversifying material supply sources enhances competitiveness.Challenges in technology adoption is addressed by investing in employee training and forming strategic partnerships with established tech companies and research institutions.Our brand-building strategy centres on targeted marketing campaigns,emphasizing product quality,unique features,and sustainability to create a strong and impactful brand identity.

Threats - To counter economic fluctuations,we diversify our supplier base and closely monitor operational costs to minimize waste.Combatting market awareness challenges involves providing product samples to potential customers and collaborating with renowned distributors and retailers for broader reach.In the face of competition,our strategy involves product differentiation based on quality and sustainability,coupled with continuous innovation to align with evolving customer preferences.

Risk assessment

Risk/impact description	Likelihood	What we'll do to reduce this risk
Having another wave of COVID19 pandemic	Highly likely	Having prepared with the required materials and strengthening the COD and online customer services.
Insufficient Financing	Likely	Obtaining government-supported green loans up to 30Mn LKR
Risk of economic crisis	Unlikely	Having a diverse segmentation such as the hotel industry customers,eco-friendly organisations, etc. and having a proactive financial management
Changes in the regulatory framework for startup businesses	Unlikely	Being updated on such compliances while meeting the ongoing regulations such as environmental standards.
Negative changes of the consumers' perception about G2FS	Unlikely	Promote transparent communication to the customers about the benefits of the products including the impact for the environment

Laws you will need to comply with

- Register the business name - Company Act No. 7 of 2007.
- Register as a limited liability company - Companies Act No. 17 of 1982.
- Obtain approval from the Board of Investment of Sri Lanka - BOI Law No. 4 of 1978.
- For exports - Sri Lanka Export Development Act, No. 40 of 1979.
- When Importing adhere to Section 47 of the Customs Ordinance.
- Resolve disputes among parties - Mediation Board Act No. 72 of 1988.
- In case of arbitration - Arbitration Act No. 11 of 1995.
- National Minimum Wage of Workers Act No. 3 of 2016.
- Budgetary Relief Allowance of Workers Act No. 4 of 2016.
- Employment of Women, Young Persons, and Children Regulations.
- EPF (Employees' Provident Fund).
- ETF (Employees' Trust Fund).
- Payment of Gratuity (Department of Labour, 2010).
- Employee Termination Act No. 45 of 1971.
- National Environmental Act (Central Bank Of Sri Lanka, 2021).
- Consumer Affairs Authority Act No. 09 of 2003 (Consumer Affairs Authority, 2024).

Goals and actions

Goals for next year

	Product designing and modification	Advocacy and education	Select suitable suppliers
Actions to achieve goal	Design a visually beautiful filter sphere self-cleaning design that requires less maintenance. Improve product reliability to use for 500 – 1000 wash cycles. Test the product for 5 months before start manufacturing. Finish the manufacturing within 1 year time	Educate people regarding the impact of this filter sphere and effects of the microfiber contamination. Select 10 areas for the awareness per week. Prepare booklets Prioritize the people who are concerned about sustainable products. Finish awareness program within 5 months.	Obtaining environmentally appropriate materials and create a trustworthy supply chain. Within first 2 months evaluate potential suppliers. Prepare proposals for suppliers over 1 month. Give more attention to suppliers who supply environmentally appropriate materials. Sign agreements with suppliers within 6 months.
Due date	4/02/2025	10/07/2025	22/08/2025
Who's responsible	Production manager	Director board and staff	Procurement manager

Goals for the next 3 years

	Environmental sustainability	**Product innovation**	**Market expansion**
Actions to achieve goal	Reduce microfibre pollution in oceans. Evaluate how much microfibers capture in 1 laundry cycle. Maximise the ability to capture microfiber within 1 laundry cycle. Use sustainable materials to ensure environmental sustainability. Improve the sustainable effects within 1 year.	Improve the product with user replaceable parts. Research and development with technology partners. This will result in using the product more than 1000 laundry cycles.	Expand the current market and increase market share. Identify 5 different international markets in 3 months. Give proposals for them over 3 months. Mainly target laundry sectors. Finalise the newly selected markets within next 3 years.
Due date	8/04/2026	12/08/2027	6/07/2028
Who's responsible	Quality controllers	Production manager	Marketing manager

The finances

Start-up costs

The G2FS with a detailed budget plan for initial start-up costs amounting to **LKR 34,500,000,**prioritizes Research & Development,Product Design,Testing & Prototyping, to emphasize innovation.Our manufacturing setup integrates cutting-edge equipment and Logistics,facilities,compliance,marketing, and a robust employee onboarding structure to form key pillars.Offices,licenses,partnership development, and contingency planning are carefully addressed.

(Refer Appendix A)

Sources of funding

Our financial blueprint is anchored with two basic pillars:

1. **Green Loans**

- We are engaged to secure a substantial investment of LKR 30 million as per Commercial/DFCC banks as per a fixed interest rate of 15%.

2. **Founding Team's Personal Investments as Capital**

- The founding team is planning to collectively invest LKR 2,000,000 of personal funds into the project, demonstrating a testament to our confidence in the project's viability and underscores our personal belief in the transformative potential of the Green Fiber Filter Sphere.

Total Funding Secured = LKR Rs. 32 million

The Ask

The Financial Snapshot:

Total Funding Secured: LKR 32,000,000.

Total Start-up Costs: LKR 34,500,000.

Net Start-up Costs After Funding: LKR 2,500,000.

In our pursuit of operational excellence and market leadership, we are actively seeking an investment of LKR 5 million to fortify the startup phase of our business. This capital infusion will play a pivotal role to ensure the successful establishment of the Green Fiber Filter Sphere as a formidable player in the eco-friendly filtration solutions market.

This calculated financial approach is finalized by a net margin for contingency, ensuring not only the successful initiation of our venture but also the resilience to navigate unforeseen challenges.

Start-up funding

	Pre-seed stage	Seed stage	Series
Product	**Research & Development,** **Product Design,** **Packaging Design,** **Licensing and certifying**	**Product Testing and Prototyping,** **Market Testing** **Initial Production set up,** **Initial Logistics set up,** **Product Post certification,** **Partnership Development,**	**Branding and Marketing,** **Administrative workplace set up,** **Product Improvement,** **Partnership developments,** **Employee Recruitment,**

Team Size	4	15	30
Monthly revenue	-	-	Rs. 46,200,000
Net profit/loss	-	(Rs.34,483,000)	Rs. 10,385,120
Equity sold	-	Rs. 2,000,000	Rs. 7,000,000

Revenue projection

(Refer Appendix B)

Annual Forecasted Revenue Projection

	Rate	Total LKR (Rs)	Month 1 LKR (Rs)	Month 2 LKR (Rs)	Month 3 LKR (Rs)	Month 4 LKR (Rs)	Month 5 LKR (Rs)	Month 6 LKR (Rs)	Month 7 LKR (Rs)	Month 8 LKR (Rs)	Month 9 LKR (Rs)	Month 10 LKR (Rs)	Month 11 LKR (Rs)	Month 12 LKR (Rs)
CASH INFLOWS														
International Sales		453,600,000	37,800,000	37,800,000	37,800,000	37,800,000	37,800,000	37,800,000	37,800,000	37,800,000	37,800,000	37,800,000	37,800,000	37,800,000
Local Sales		100,800,000	8,400,000	8,400,000	8,400,000	8,400,000	8,400,000	8,400,000	8,400,000	8,400,000	8,400,000	8,400,000	8,400,000	8,400,000
TOTAL INFLOWS		**554,400,000**	**46,200,000**	**46,200,000**	**46,200,000**	**46,200,000**	**46,200,000**	**46,200,000**	**46,200,000**	**46,200,000**	**46,200,000**	**46,200,000**	**46,200,000**	**46,200,000**
RECURRENT EXPENDITURE														
Materials														
Product Components		372,614,880	31,051,240	31,051,240	31,051,240	31,051,240	31,051,240	31,051,240	31,051,240	31,051,240	31,051,240	31,051,240	31,051,240	31,051,240
Packaging														
Primary Packaging requirements (Box, Printing, Bubble wrap)		7,920,000	660,000	660,000	660,000	660,000	660,000	660,000	660,000	660,000	660,000	660,000	660,000	660,000
Secondary Packaging Requirements		420,000	35,000	35,000	35,000	35,000	35,000	35,000	35,000	35,000	35,000	35,000	35,000	35,000
Labour costs														
Plant Workers														
Quality Controllers (Sourcing/Final)		883,200	73,600	73,600	73,600	73,600	73,600	73,600	73,600	73,600	73,600	73,600	73,600	73,600
Production Labourers		1,159,200	96,600	96,600	96,600	96,600	96,600	96,600	96,600	96,600	96,600	96,600	96,600	96,600
Packaging Helpers		2,649,600	220,800	220,800	220,800	220,800	220,800	220,800	220,800	220,800	220,800	220,800	220,800	220,800
Administrative Workers														
Production Manager		496,800	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400
Human Resources Manager		496,800	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400
Finance Manager		496,800	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400
Procurement Manager		496,800	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400
Marketing Manager		496,800	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400
Logistics Manager		496,800	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400
Customer Relationship Manager		496,800	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400	41,400
Other														
Security Guards		662,400	55,200	55,200	55,200	55,200	55,200	55,200	55,200	55,200	55,200	55,200	55,200	55,200
Cleaning Staff		441,600	36,800	36,800	36,800	36,800	36,800	36,800	36,800	36,800	36,800	36,800	36,800	36,800
Labour On-costs														
Superannuation (EPF/ETF)		1,630,080	135,840	135,840	135,840	135,840	135,840	135,840	135,840	135,840	135,840	135,840	135,840	135,840
Equipment maintenance		75,000	75,000	-	-	-	-	-	-	-	-	-	-	-
Occupancy costs														
Manufacturing Plant Rent		1,800,000	150,000	150,000	150,000	150,000	150,000	150,000	150,000	150,000	150,000	150,000	150,000	150,000
Office Space Rent		900,000	75,000	75,000	75,000	75,000	75,000	75,000	75,000	75,000	75,000	75,000	75,000	75,000
Logistics Costs		1,800,000	150,000	150,000	150,000	150,000	150,000	150,000	150,000	150,000	150,000	150,000	150,000	150,000

Communications and systems														
telephone		60,000	5,000	5,000	5,000	5,000	5,000	5,000	5,000	5,000	5,000	5,000	5,000	5,000
internet costs		120,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000
Administration														
stationery and office requisites		120,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000
legal expenses		300,000	25,000	25,000	25,000	25,000	25,000	25,000	25,000	25,000	25,000	25,000	25,000	25,000
Utility Expenditure (Electricity and Water)		600,000	50,000	50,000	50,000	50,000	50,000	50,000	50,000	50,000	50,000	50,000	50,000	50,000
Other Expenses;														
Audits (Sustainability & Accounting)		440,000	220,000	-	-	-	-	-	220,000	-	-	-	-	-
Interests payable		4,500,000	375,000	375,000	375,000	375,000	375,000	375,000	375,000	375,000	375,000	375,000	375,000	375,000
Post Certification (SLSI)		2,000,000	2,000,000	-	-	-	-	-	-	-	-	-	-	-
TOTAL RECURRENT EXPENDITURE		404,753,560	35,814,880	33,519,880	33,519,880	33,519,880	33,519,880	33,519,880	33,739,880	33,519,880	33,519,880	33,519,880	33,519,880	33,519,880
NET INCOME / CLOSING CASH		149,646,440	10,385,120	12,680,120	12,680,120	12,680,120	12,680,120	12,680,120	12,460,120	12,680,120	12,680,120	12,680,120	12,680,120	12,680,120

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Appendices

Appendix A

Budget Plan for Initial Start-up costs	
Once-off expenditures	**LKR**
Research & Development	
Product Development & Design	15,000
Testing & Prototyping	50,000
Manufacturing Setup	
Equipment Rent & Set up Expenditure	
Embedding Machinery	715,000
Initial Material Inventory	
Product Components	31,053,000
Packaging Requirements	695,000
Distribution & Logistics	
Intial Shipping Set up	150,000
Facility Rent related Expenditure	
Office, Manufacturing Space & Warehouse	250,000
Regulatory Compliances & Certifcation Costs	25,000
Marketing and Branding	
Website Development	30,000
Initial Marketing & Advertising campaigns	30,000
Employee Onboarding Expenditure	
Employee Recruitment & Training	20,000
Initial Employee salary expenses	910,000
Office Set up	
Office Computers and Machinery	250,000
Initial Office Stationary and Requisites	10,000
Miscellaneous Expenditure	
Licenses	30,000
Partnership Development costs	150,000
Contingency requirements	100,000
NET START-UP COSTS	**34,483,000**

Appendix B

Sales = International Sales + Local sales

(9000*4200) + (2000*4200)

LKR 46,200,000

Material Expenditure = Cost of inner and outer components

(Nano Fiber material + Multi filtrations layers + Anti-Microbial Coating + Smart Sensor Technology + Automated Self-cleaning Mechanism + Outer appearance materials)

= ($1.50 + 2.00 + 0.50 + 1.50 + 1.50 + 1.50 + 0.25) *322.61

= 8.275*322.61

= 2822.84 * 11,000(Units)

= **LKR 31,051,240**

Packaging Materials = Primary Packaging + Secondary packaging required for shipping

= (Box , Bubble wrap & Printing) + (Box)

= (60*11,000) + (100*350)

= 660,000 + 35,000

= **LKR 695,000**

Labour Costs

(According to Sri Lanka Law it if required for the employee to pay a rate of 8% of their monthly salary as per superannuation. Employers must pay 12% of the salary of an employee as per superannuation. In total a 20% rate is excused as per payment of superannuation)

Plant Worker costs = ((Quality Controller Salary * No.of workers)-Superannuation rate) +

((Production Labour Salary * No.of Workers)– Superannuation rate) +

(Packaging Helper Salary* No.of Workers – Superannuation rate)

= ((40,000*2)-8%) + ((35,000*3)-8%) + ((30,000*8)-8%)

= 73,600 + 96,600 + 220,800

= **LKR 391,000**

Administrative Staff costs = (Administrative Staff Salaries - Superannuation rate)

= ((45,000*7)-8%)

= **LKR 289,800**

Plant Worker costs = ((Security Guards Salary * No.of workers)-Superannuation rate) + ((Cleaning Labour Salary * No.of Workers)– Superannuation rate)

= ((30,000*2)-8%) + ((20,000*3)-8%)

= 55,200 + 36,800

= **LKR 92,000**

Labour On-Costs = Superannuation Costs

= (Total allocated Employee Salaries * 20%)

= ((80,000 + 105,000 + 240,000) + (45,000*7) + (60,000 + 40,000)) * 20%

= **LKR 135,840**

Appendix C

BREAK EVEN ANALYSIS

Production Capacity (per Month)			**11,000**
		per Unit	per Month
Fixed Costs			
1	Salary Expenditure	32.07	352,800
2	Rental Expenditure	20.45	225,000
3	Utilities	4.55	50,000
4	Accounting and legal	2.73	30,000
5	Telephone	0.45	5,000
6	Internet	0.91	10,000
7	Interest	34.09	375,000
8	Security	6.11	67,200
9	Cleaning Facilities	4.07	44,800
	Total Fixed Cost	**105.44**	**1,159,800**
Variable Cost			
1	Materials (Product Supplies)	**2,822.84**	31,051,213
2	Logistics Costs	**13.64**	150,000
3	Packaging Materials	**63.18**	695,000
4	Direct Labour	**29.75**	327,240
	Total Variable Cost	**2,929.40**	**32,223,453**
Targeted Selling Price per Unit		**4,200**	
	Break Even Point (units)		**913** units
	Break Even Point (revenue)	**3833762**	

Appendix D

Minimum Viable Product that shows the 3 multi-filtration layers, which makes the product differentiated from competitors (Barringer & Ireland, 2022).

